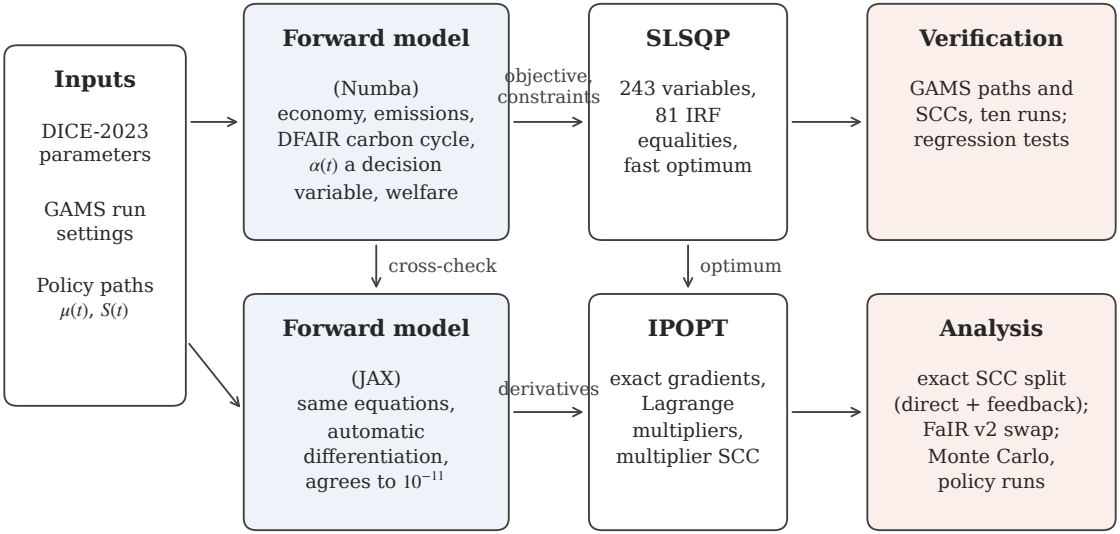
Inputs
DICE-2023 parameters
GAMS run settings
Policy paths $\mu(t)$, $S(t)$
Forward model
(Numba) economy, emissions, DFAIR carbon cycle, $\alpha(t)$ a decision variable, welfare
objective, constraints
SLSQP
243 variables, 81 IRF equalities, fast optimum
Verification
GAMS paths and SCCs, ten runs; regression tests
cross-check
optimum
Forward model
(JAX) same equations, automatic differentiation, agrees to 10^{-11}
derivatives
IPOPT
exact gradients, Lagrange multipliers, multiplier SCC
Analysis
exact SCC split (direct + feedback); FaIR v2 swap; Monte Carlo, policy runs